

Simultaneity Bias

Supply:

$$Q = \alpha_1 * P + \alpha_2 z_1 + \delta_t$$

Demand:

$$Q = \beta_1 * P + \beta_2 z_2 + \epsilon_t$$

Two equations two unknowns:

So:

$$0 = \alpha_1 * P_t + \alpha_2 z_1 + \delta_t - (\beta_1 * P_t + \beta_2 z_2 + \epsilon_t)$$

$$-\alpha_1 * P_t + \beta_1 * P_t = \alpha_2 z_1 + \delta_t - (\beta_2 z_2 + \epsilon_t)$$

$$P_t(-\alpha_1 + \beta_1) = \alpha_2 z_1 + \delta_t - (\beta_2 z_2 + \epsilon_t)$$

$$P_t = \frac{\alpha_2 z_1 + \delta_t - (\beta_2 z_2 + \epsilon_t)}{\beta_1 - \alpha_1}$$

$$P_t = \frac{\alpha_2}{\beta_1 - \alpha_1} z_1 - \frac{\beta_2}{\beta_1 - \alpha_1} z_2 + \frac{(\delta_t - \epsilon_t)}{\beta_1 - \alpha_1}$$

$$Q_t = \alpha_1 * \left(\frac{\alpha_2}{\beta_1 - \alpha_1} z_1 - \frac{\beta_2}{\beta_1 - \alpha_1} z_2 + \frac{\delta_t - \epsilon_t}{\beta_1 - \alpha_1} \right) + \alpha_2 z_1 + \delta_t$$

$$Q_t = \alpha_2 \left(1 + \frac{\alpha_1}{\beta_1 - \alpha_1} \right) z_1 - \frac{\alpha_1 \beta_2}{\beta_1 - \alpha_1} z_2 + \frac{\alpha_1 * (\delta_t - \epsilon_t) + (\beta_1 - \alpha_1) \delta_t}{\beta_1 - \alpha_1}$$

$$Q_t = \alpha_2 \left(1 + \frac{\alpha_1}{\beta_1 - \alpha_1} \right) z_1 - \frac{\alpha_1 \beta_2}{\beta_1 - \alpha_1} z_2 + \frac{\beta_1 \delta_t}{\beta_1 - \alpha_1} - \frac{\alpha_1 \epsilon}{\beta_1 - \alpha_1}$$